



Cloud-Native Bookstore

Milestone 2: Infrastructure Validation

Group 5 | May 7, 2026

Agenda

1. **Project Status:** Current deployment state
2. **Core Deliverables:** Pod status & Architecture
3. **Backend API:** Health probes & Business logic
4. **Security & Routing:** NetworkPolicy & Service mesh
5. **Summary:** Achievements & Next steps



1. Project Status: Verified Deployment



Current Cluster State

- **Backend**: 2 replicas running (1/1 Ready)
- **Frontend**: 1 replica running (1/1 Ready)
- **Zero Restarts**: Stable deployment since creation
- **Environment**: Minikube (Local Validation)



Reproducible Workflow

- **Strategy**: Kustomize overlays for environment separation
- **Command**: `kubectl apply -k k8s/overlays/minikube`
- **Benefit**: One-click deploy for Dev & Production



2. Core Deliverables: Live Cluster View

Real-time Pod Status

```
$ kubectl get pods -n bookstore -o wide
```

NAME	READY	STATUS	AGE
bookstore-backend-xxxx	1/1	Running	9m
bookstore-backend-xxxx	1/1	Running	9m
bookstore-frontend-xxxx	1/1	Running	19m

Infrastructure Architecture

-  **Backend (2x)**: Python API, Port 8000
-  **Frontend (1x)**: Nginx UI, Port 80 → NodePort 30080
-  **Data Layer**: Persistent Volume Claim (Postgres)
-  **Security**: Zero-trust Network Policies

3a. Backend API: Health Probes

Kubernetes Probe Configuration

We configure automated health checks to ensure high availability.

```
# deployment-backend.yaml
livenessProbe:
  httpGet:
    path: /healthz
    port: 8000
readinessProbe:
  httpGet:
    path: /ready
    port: 8000
```

Benefits

- **Auto-healing**: Restarts stuck containers automatically
- **Traffic Control**: Stops routing traffic to unready pods
- **Zero Downtime**: Ensures safe rolling updates

3b. Backend API: Business Logic

Cluster-Internal Verification

We verified the API logic returns real data using internal routing.

```
$ kubectl exec -n bookstore <backend-pod> -- python3 -c "..."
```

Verified Output

```
{
  "count": 3,
  "books": [
    {"title": "Cloud Native Patterns"},
    {"title": "Kubernetes in Action"},
    {"title": "Designing Data-Intensive Applications"}
  ]
}
```

“ **Conclusion:** Business logic implemented and returning structured JSON.”

4. Security & Routing: Enforced Policies

Non-Root Execution (Container Hardening)

Service	User	UID	Security Context
Backend	appuser	1000	runAsNonRoot, drop: ["ALL"]
Frontend	nginx	101	runAsUser: 101, drop: ["ALL"]

Zero-Trust Network Policies

-  **Default Deny**: No traffic allowed unless explicitly permitted
-  **Database**: Accessible **only** from backend pods (Port 5432)
-  **Routing**: Backend allowed to proxy to Frontend (Port 80)
-  **External**: Restricted to NodePort 30080

5. Summary & Next Steps

Milestone 2 Achievements

1.  **Infrastructure**: 9 K8s resources deployed & stable
2.  **Security**: Non-root execution + NetworkPolicy enforced
3.  **Observability**: Health probes active & responding
4.  **Routing**: Service mesh validated via ClusterIP
5.  **Logic**: API returning real book data

Next Steps

- Finalize external access (Ingress / Tunnel)
- Implement CI/CD pipeline (GitHub Actions)
- Record end-to-end demo video

Q & A

Thank you!

